

KRISHAN AI

Detailed Project Report

Farm-level climate-risk intelligence and decision support for Bangladesh

Competition: Grameenphone Future Makers — AI for Social Good

Focus: Environmental Sustainability & Climate Resilience

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Originality • Problem relevance • AI integration • Impact • Feasibility • Market sustainability

September 2026 | Version 1.0 | Concept/prototype-stage proposal. Planned capabilities are not presented as implemented or validated.

02. Executive Summary

Purpose and central idea

Krishan AI is a proposed agricultural climate-intelligence and decision-support platform. It aims to connect fragmented environmental information to a specific farm's context, explain a risk assessment, communicate uncertainty and link the result to a practical next step. Its central value is not simply showing data; it is helping users interpret data for decisions.

The long-term vision includes soil salinity, flooding and waterlogging, drought and water stress, and weather-crop risks. The first demonstrator must be narrowed to one measurable use case after data discovery. Salinity is a candidate starting point, not a claim that every module already exists.

Credibility principle

The team will inventory accessible data, consult farmers and practitioners, define a target and horizon, establish baselines, compare appropriate model candidates and validate before deployment. If evidence cannot support farm-level precision, Krishan will narrow the claim or collect more data.

Current status

This is a concept/prototype-stage proposal. No field deployment, validated accuracy, confirmed partnership, proven uptake or measured loss reduction is claimed. Costs, sample sizes and dates are planning assumptions pending verification.

03. Problem Statement And Relevance

The problem

Farmers may receive a weather forecast, soil reading, regional risk map or extension advice yet still lack an integrated answer for their own field: what risk is developing, what evidence supports it, how uncertain is it, and what action may fit the crop and growth stage? Data are collected at different scales, formats and times. The challenge is turning them into a usable decision.

Salinity illustrates this gap. Soil and irrigation-water salinity, rainfall, temperature, soil moisture, location and crop conditions can interact. One reading is a snapshot; a regional forecast may not represent each field. Late or fragmented information can limit preparation.

Why it matters

Climate-related risks can affect planting, irrigation, input timing, crop stress and farm income. Smallholders may not be able to afford sensors, repeated lab tests or specialist interpretation. A service must be understandable, affordable, low-bandwidth and honest about uncertainty.

Evidence discipline

This report does not claim comparable services do not exist. Bangladesh has agricultural advisories, crop-zoning information and regional monitoring. Krishan's hypothesis is that a farm-context workflow can fill a demonstrated gap; this must be tested with users and benchmarked against existing tools.

04. Target Groups And User Needs

Primary users

The primary target group is smallholder farmers exposed to climate-related agricultural risks. The service must accommodate varied literacy, smartphone access, connectivity, resources and familiarity with technical data. Bengali-first communication and a non-smartphone route should be core requirements.

Supporting users

- Extension officers and field agents who collect observations and follow up on alerts.
- Agronomists who review target definitions, recommendation rules and uncertain cases.
- NGOs, cooperatives and agricultural programmes serving multiple farms.
- Potential institutional users such as agencies, insurers or agribusinesses, subject to demonstrated value and written agreements.

Jobs to be done

Farmers need to understand whether risk is rising, why the system says so, what is unknown and what safe next step may be appropriate. Officers need to prioritise follow-up. Experts need traceable evidence and override controls. Institutions may need aggregated information without unnecessary exposure of personal data.

During the first 4–6 months, interview farmers and practitioners where feasible about current decisions, trusted channels, affordability, comprehension and service gaps. Design should follow findings rather than assume everyone wants an app.

05. Proposed Ai-Enabled Solution

Workflow

Farm registration → data collection and quality checks → risk estimation → explanation and uncertainty display → expert-reviewed decision support → farmer communication → feedback and outcome capture. Each stage needs a quality check so unreliable inputs or unsafe advice do not pass through silently.

Farm Context Model

A structured profile may include farm location/boundary, crop, planting date, growth stage, soil and water readings, weather, satellite-derived indicators and available history. This is not a claim that a full digital twin exists. A simulation-capable Farm Digital Twin is a possible future direction.

Climate Risk Engine

The engine will estimate a defined target over a stated time horizon and geographic unit. Each output should include a risk category, supporting factors, data freshness, uncertainty/confidence information and data-sufficiency status—not just an unexplained score.

Decision support and delivery

A separate recommendation layer will match expert-reviewed rules to risk, crop, growth stage and context. Bengali SMS, voice, an optional app or officer-assisted delivery can be tested. Generative AI, if used, may simplify approved guidance but must not invent agricultural instructions.

06. Specific Role Of Ai And Model Research

Where AI may help

Machine learning may identify patterns across multiple variables and estimate risk where simple rules are insufficient. It can support prediction, pattern recognition, risk triage and decision support. Data validation, geospatial joins, access control and approved recommendation rules may be better handled with conventional software.

Research-first selection

First define the target, prediction horizon, spatial unit, crop/season and cost of errors. Assess labels, sample size, missingness, coverage and leakage. Build threshold/statistical baselines first. Regression, tree-based methods, neural networks or image models are candidates only if data structure and evidence justify them.

Validation and explanation

Use held-out data and, where possible, separate farms, locations and later periods. Evaluate class-specific precision/recall, false alarms, missed high-risk cases, lead time and calibration where confidence is shown. Explanations identify factors associated with predictions, not proven causes.

Uncertainty

Confidence values must be tested. If inputs are stale, missing or outside validated conditions, show “insufficient data”, request another reading or escalate to human review. Keep a model only if it produces a repeatable improvement over a baseline.

07. Data Strategy And Accessibility

Sources to investigate

- Public satellite products such as Sentinel-1/2 and Landsat, subject to suitability and processing capability.
- Weather/climate APIs; national station archives may require permission or agreement.
- Soil and water observations, including EC for salinity-related research.
- Crop, planting, growth-stage, management and outcome records collected with consent.
- River, tide, groundwater and water-management data where access and relevance can be confirmed.

Free does not mean sufficient

Satellite data can be free but require geospatial expertise; optical observations may be cloud-limited. Weather data may be regional rather than farm-scale. Soil/crop records may be incomplete or restricted. Repeated ground truth is likely to be the main bottleneck for reliable farm-level predictions.

Data inventory

For every source record owner, access method, permissions, spatial/temporal resolution, update frequency, historical depth, missingness, units, bias, cost and intended use. Mark it confirmed, under investigation or unavailable. If access fails, use an alternative or narrow the product claim.

08. Ground Truth And Field Collection

Why field evidence matters

Public data support early exploration, but farm-level claims need observations matching the prediction target. For salinity, repeated EC readings and water-condition records may be needed alongside crop, soil, rainfall and location. Ground truth may require calibrated meters, lab comparisons or expert assessment.

Proposed protocol

- Register farms with consent and standardised location/crop fields.
- Use consistent measurement points, depths, units and time records.
- Calibrate instruments; log calibration and take duplicate readings where practical.
- Record instrument, observer, weather context and unusual conditions.
- Document missing readings rather than silently imputing them.
- Link measurements to outcomes only when time and location alignment is defensible.

Sampling and governance

A 100-farm pilot is a proposal, not a guarantee of statistical power or national generalisation. Sampling depends on hazard variability, event frequency, crop diversity, cost and intended claims. Farmers should understand what is collected, why, who accesses it, retention and withdrawal options. Separate identifiers from analysis records and aggregate reporting where possible.

09. System Architecture

Proposed layers

1. Data sources. 2. Ingestion and quality checks. 3. Farm Context Model. 4. Climate Risk Engine. 5. Decision support. 6. Farmer/expert communication. 7. Monitoring, audit and feedback. Data flows through the layers; field outcomes and corrections feed later evaluation.

Design principles

- Keep the first stack small and modular; avoid complexity before validating the use case.
- Use geospatial database features if farm boundaries and spatial joins are required.
- Separate farmer, officer, expert and administrator permissions.
- Support intermittent connectivity with queued uploads, delayed sync and SMS/voice fallback.
- Log data freshness, model version, sufficiency, rule version and delivery status.

Minimum viable loop

An early prototype can use a farm registry, auditable data pipeline, baseline risk logic, clear output screen and curated recommendation library. Dense IoT, a large language model, nanotechnology and a digital twin are not prerequisites.

If a source is stale, a sensor fails or a prediction falls outside validated conditions, mark the limitation and use a safe fallback. Never invent missing values or present old predictions as current.

10. Hardware And Field Operations

Hardware philosophy

Introduce hardware only when it resolves a data gap that public sources and periodic field measurements cannot address. Start with data discovery and software baselines. If sensing is justified, deploy a small representative network before buying equipment for every farm.

Potential equipment

- Handheld EC meter and calibration solutions for periodic soil/water readings.
- Limited soil moisture/EC probes and loggers at sentinel farms, if justified.
- GPS-enabled phones/tablets for field records.
- Shared weather instruments or community gateways if local coverage requires them.
- Enclosures, mounting, power, spares and a repair/calibration plan.

Shared hubs and maintenance

Community gateways may reduce connectivity costs across several farms, but range, terrain, power and maintenance must be tested. Meters and probes need cleaning, calibration, replacement and field visits; monsoon flooding, silt and corrosion can interrupt observations. Nanomaterial sensors and advanced nutrient sensing are future research options, not initial requirements.

11. Prototype And Farmer Experience

Prototype objective

Demonstrate the decision workflow without implying every model already works. A user should register/select a farm, view evidence and data freshness, see a clearly labelled demo/research-stage output, understand contributing factors and view an approved action pathway.

Illustrative message only

“Risk: High; confidence: Moderate; data status: recent soil reading missing. Evidence considered: recent rainfall and last recorded EC reading. Next step: request an updated measurement or contact the field officer.” This is a format example, not a real prediction or farm-specific recommendation.

Accessible communication

Use plain Bengali and explain technical terms. Test comprehension. SMS should be concise; voice may help users who prefer audio. An app should be optional. Distinguish urgent alerts from general information and cases requiring an officer.

Expert interface

Experts should inspect input quality, model/rule versions and uncertainty, override or suspend recommendations, and record decisions. User testing should assess comprehension, feasibility given money/labour/water, trust and the route to question a surprising alert.

12. Feasibility And Roadmap

Months 0-4/6: discovery

Explore user needs across regions; inventory datasets; assess access, resolution and quality; consult farmers and experts where possible; compare candidate crops/risks; define ground truth, target and prediction horizon. Deliverables: data register, user-needs summary, initial scope decision, validation protocol and risk register.

Months 6-12: data and model research

Clean/align accessible data, build the Farm Context Model, establish baselines, compare candidates and assess whether farm-level claims are supportable. Prototype explanations, uncertainty and approved recommendation delivery. An MVP around month 12 is conditional on readiness, not guaranteed.

Years 1-2: controlled pilot

Test the workflow with a sample designed for the chosen risk and geography. Measure performance, false alarms, missed warnings, lead time, comprehension, uptake, cost per farm and workload. Revise or narrow scope if results are inadequate.

Years 2-3: staged expansion

Expand within the validated context, then to new areas and risks only when local data, ground truth, expert review and capacity are sufficient. Nationwide coverage remains a long-term ambition.

13. Cost Model And Budget Transparency

Caveat

Figures below are planning estimates from the supplied technical feasibility draft, not supplier quotes or committed funding. Bangladesh costs depend on sourcing, import duties, taxes, specifications, exchange rates, labour, travel, calibration and connectivity. Obtain local quotations before procurement.

Indicative hardware estimates

Field sensor node with moisture/EC, solar and LoRa: USD 40–70 per unit; community gateway: USD 120–200; enclosure/mounting/surge protection: USD 10–20 per node. These may exclude installation, calibration, spares, duties, visits and replacement.

Indicative monthly estimates in the draft

For a proposed pilot around 1,000 farmers and 60 hubs: cloud/database/storage USD 150–400/month; SMS USD 50–300/month; cellular data USD 2–5 per hub/month; model serving USD 100–300/month. These assumptions need verification and do not make a full project budget.

Costs still to calculate

- Team time; field visits and travel; lab testing and calibration.
- Hosting, messaging, monitoring, backups and security.
- Maintenance, replacement devices, repairs and training.
- Consent, independent evaluation, documentation and user support.

Avoid instrumenting every farm before showing hardware improves results enough to justify lifetime cost. Track cost per farm and per useful warning, not only purchase price.

14. Differentiation And Originality

Positioning

Krishan is proposed as a farm-level climate-risk interpretation and decision-support layer—not simply a weather app, sensor, satellite dashboard or general chatbot. The intended distinction is the workflow from farm context and multi-source evidence to risk, uncertainty, reviewed action and accessible delivery.

Potential differentiators

- Farm context: account for location, crop, growth stage and soil/water evidence.
- Data sufficiency: say when evidence is too weak for a reliable prediction.
- Decision connection: map outputs to expert-approved, context-aware recommendations.
- Accountability: human review, versioned rules, override and incident correction.
- Access: Bengali-first communication and non-smartphone routes.
- Modular expansion: add hazards only when data and validation are ready.

Benchmark rather than assert

Compare government advisories, apps, regional hazard products, research prototypes, sensor services and NGO programmes. Assess resolution, update frequency, hazards, guidance, transparency, validation, channel and cost. If a similar service exists, define a complementary role rather than claiming no competitor exists. Potential defensibility may grow from consent-based local data, trusted workflows and validated rules, but these are assets to earn.

15. Market Survival And Sustainability

Who benefits and who pays

Farmers are primary beneficiaries, but many may not pay for a new service before its value is proven. A mixed financing model may be more realistic to test than individual subscriptions alone. Potential institutional customers include agricultural programmes, NGOs, cooperatives, insurers, agribusinesses and public agencies, subject to demand and formal agreements.

Possible funding/revenue

- Grants, competitions and research support for discovery and validation.
- Institutional contracts for aggregated dashboards and programme reporting.
- Sponsored/subsidised alerts through agricultural or community programmes.
- Optional paid organisational services after measurable value is shown.
- Low-cost farmer tier only if research supports ability and willingness to pay.

Why it could survive

The service must solve a real decision problem, earn trust, perform under field conditions, keep maintenance affordable and find repeat institutional demand. Alerts that do not change decisions will be ignored; unreliable sensors damage trust; free existing services may reduce willingness to pay; institutional procurement may be slow. Test buyer interest, cost to serve, renewal intent and operational savings before scaling.

16. Expected Social And Environmental Impact

Potential benefits

If predictions are reliable and guidance actionable, Krishan could provide earlier awareness of risks, improve understanding of field conditions and help farmers time irrigation or crop decisions. Agricultural programmes may use aggregated data to prioritise visits. These are hypotheses to test, not guaranteed outcomes.

Measurement framework

- Predictive: class-specific performance, calibration, false alarms, missed warnings and lead time.
- User: comprehension, trust, usefulness, accessibility and recommendation uptake.
- Farm: management changes, input use, crop stress and yield outcomes where measurable.
- Equity: participation across farm size, gender, tenancy, geography and connectivity.
- Operations: cost per farm/alert, response time and staff workload.

Evaluation and safeguards

Where feasible, compare outcomes with a suitable baseline or comparison group and record confounders. A small pilot cannot prove nationwide impact. Report sample, period, missingness and limitations. Provide consent and complaint routes, non-digital access, and safeguards against using risk labels to deny credit, insurance or support.

17. Responsible Ai, Privacy And Safety

Principle

AI outputs are decision support, not authority. High-impact actions require approved agricultural guidance and human review. The system should be transparent, contestable and proportionate to evidence.

Safeguards

- Minimise collection and explain purpose, sharing, retention and withdrawal options.
- Restrict access, protect data in transit/at rest and maintain audit logs.
- Flag stale, missing, inconsistent or out-of-range observations.
- Avoid confident predictions outside validated conditions.
- Version models/rules and enable rollback or suspension.
- Constrain generative AI to approved content and test Bengali wording.
- Provide a route to report incorrect alerts and document corrections.

Fairness and incident handling

Performance may vary by region, soil, crop, season and sensor access. Monitor differences and do not silently extend models to unvalidated contexts. Include women farmers, tenant farmers and remote communities in research. For possible harm, record the case, investigate, suspend problematic rules if needed and notify affected users where appropriate.

18. Risk Register And Mitigation

Technical/data risks

- Sparse ground truth: narrow target, start with baselines and collect repeated representative measurements.
- Satellite gaps/resolution: use imagery as context, not sole evidence for plot-level claims.
- Weather resolution mismatch: do not imply farm precision without validated downscaling.
- Sensor drift/failure: calibration, maintenance and sufficiency flags.
- Model drift: monitor and revalidate before expansion.

Operational and market risks

- Connectivity gaps: SMS/voice, officer support and queued uploads.
- Advice not feasible: co-design with farmers and agronomists.
- Partner delays: begin with accessible public data where possible.
- Hardware over budget: obtain local quotes and compare with manual sampling.
- Weak willingness to pay: test institutional and subsidy models.

Go/no-go rule

Pause or narrow if data are inadequate, the model does not beat baseline, false negatives are unacceptable, expert review is unavailable or cost to serve is unsustainable. A documented decision not to deploy is better than an unsafe product.

19. Validation And Success Criteria

Define the task

Specify target variable, spatial unit, prediction horizon, crop/season, intended decision and cost of errors. For salinity, define how labels relate to EC measurements and/or observed outcomes with agronomic review. A generic “climate risk score” is too vague to validate.

Baselines and testing

Compare candidate models with thresholds, historical averages or other defensible baselines. Prevent leakage by separating related observations where necessary. Use held-out data and, where possible, separate locations and later periods.

Pilot measures

- Precision/recall by risk class and calibration if confidence is shown.
- False alarms, missed high-risk cases and warning lead time.
- Data sufficiency, stale-input rate and withheld predictions.
- Farmer comprehension, trust, usefulness and uptake.
- Cost per farm, cost per alert and operational workload.

Expansion gate

Expansion requires repeatable performance, understandable messages, acceptable safety, sufficient data and support capacity. A 100-farm pilot is provisional and cannot establish national generalisation. Each new hazard, crop and region needs its own validation plan. Report limitations, missing data and subgroup differences alongside results.

20. Conclusion And Immediate Next Steps

Conclusion

Krishan AI proposes a bridge between environmental information and agricultural decisions. Its intended value is farm-context risk intelligence that explains evidence, communicates uncertainty and connects to reviewed guidance. The long-term vision includes multiple risks and wider reach, but credibility depends on proving one well-defined use case first.

The largest uncertainties are data access/resolution, repeated ground truth, model validity, maintenance cost, user trust and payer sustainability. Free public data can support exploration but cannot guarantee reliable farm-level prediction.

First 90 days

- Conduct structured farmer and practitioner discovery.
- Build a data register covering access, resolution, cost and quality.
- Select a candidate risk, crop, area, horizon and measurable target.
- Define ground truth and independent validation before model selection.
- Build a baseline and prototype; label demo/synthetic outputs clearly.
- Obtain local quotes for instruments, calibration, travel, hosting and messaging.
- Benchmark alternatives and interview potential institutional buyers.
- Define consent, privacy, expert-review and incident-response procedures.

Source basis and limitations

Synthesised from the team's competition Q&A, AgroShield AI Defense Report and AgroShield AI Technical Feasibility & Implementation Report. Costs, quantities, pilot size, dates, targets and partnerships are assumptions unless verified. External data access, supplier prices, market size, regulations and competitor capabilities were not independently verified for this report.